CCEM-Net: Conflict-Aware Cross Evidential Modulation for Hyperspectral Change Detection

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***Abstract*—Hyperspectral change detection (HSCD) remains challenging due to spectral inconsistency across bitemporal observations, heterogeneous spatial change patterns, and ambiguous decision boundaries that induce predictive uncertainty. Existing methods mainly rely on deterministic representation learning, lacking mechanisms to model uncertainty and resolve such conflicts, often resulting in overconfident and unreliable predictions. To address these issues, we propose CCEM-Net, a conflict-aware evidential deep learning (EDL) framework for reliable HSCD. Specifically, a change-aware interaction (CAI) module is introduced to model cross-temporal dependencies, suppress pseudo-changes, and enhance change-relevant responses via dual gating. A parallel adaptive multiscale spectral and spatial (AMSS) module further captures heterogeneous patterns through temporal-difference-guided multiscale feature aggregation. To handle decision ambiguity, we develop a conflict-aware cross evidential modulation (CCEM) mechanism that models branch predictions as Dirichlet distributions to quantify uncertainty. By measuring interbranch conflict via Jensen–Shannon (JS) divergence, CCEM adaptively modulates evidence strength. Subsequently, reliable decision-level fusion is performed using the Yager rule. Experiments on benchmark datasets demonstrate that CCEM-Net consistently achieves the superior performance in both accuracy and reliability compared with state-of-the-art (SOTA) methods. The code will be publicly available at: https://github.com/fwx0314/CCEM-Net**

***Index Terms*—Conflict-aware evidential fusion, evidential deep learning (EDL), hyperspectral change detection (HSCD), uncertainty modeling.**

I. Introduction

CHANGE detection (CD) aims to identify land-cover variations from multitemporal observations and plays a crucial role in a wide range of applications, such as environmental monitoring [1] and urban analysis [2]. With the increasing availability of hyperspectral imagery, hyperspectral change detection (HSCD) has attracted growing attention due to its ability to provide rich spectral information for fine-grained material discrimination [3], [4]. However, despite its advantages, HSCD remains a challenging task due to the high dimensionality of hyperspectral data and the complex spectral–spatial correlations [5].

Early HSCD approaches relying on statistical analysis or direct differencing, such as change vector analysis (CVA) [6], multivariate alteration detection (MAD) and its variants [7], [8], and transformation-based methods [9], measure spectral discrepancies between bitemporal observations to identify changes. However, these methods are often sensitive to noise, illumination variations, and spectral inconsistencies. Moreover, they typically rely on handcrafted features and shallow statistical assumptions, which restrict their capability to capture complex nonlinear spectral–spatial dependencies and high-level semantic variations [10]. Consequently, their robustness and generalization ability are limited when dealing with heterogeneous and large-scale real-world scenes.

Recently, deep learning has significantly advanced HSCD by enabling powerful spectral–spatial representation learning [11], [12], [13], [14]. Existing methods mainly focus on three aspects. First, feature representation enhancement has been widely explored, including multilevel feature fusion [15], cross-channel modeling [16], and spatial–spectral attention mechanisms [17], which improve discriminative capability. Second, cross-temporal interaction has been strengthened through difference-guided learning and attention-based alignment strategies, such as DIEFEN [18], AIWSEN [19], and CGMNet [20]. Third, robustness and efficiency have been improved by disentangling spectral variability [21], modeling subtle differences under noise [22], and designing efficient architectures such as Vision Mamba-based models [23].

Despite these advances, reliable HSCD remains fundamentally limited by the inconsistency of bitemporal observations across spectral and spatial levels [24], [25]. Spectral shifts caused by environmental variations often introduce pseudo-changes [26], while heterogeneous land-cover changes occur at diverse spatial scales, challenging models with fixed receptive fields [21]. More importantly, these discrepancies often induce inconsistent or even conflicting responses within spectral–spatial representations, particularly in ambiguous

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boundary regions [16], [17], [19], [27]. Such conflicts make reliable change identification inherently challenging.

However, most existing methods rely on deterministic classifiers based on softmax probabilities, which lack the ability to quantify predictive uncertainty. As a result, they tend to produce overconfident predictions even when the input data is inconsistent or ambiguous [28], [29], [30]. In practical applications, such unreliable predictions may lead to critical errors, especially in high-stakes scenarios such as disaster monitoring or environmental assessment. Therefore, beyond improving feature representation, it is essential to explicitly model uncertainty and account for conflicting evidence to achieve reliable and trustworthy HSCD.

To address this issue, evidential deep learning (EDL) has emerged as a promising framework for uncertainty-aware modeling [30]. By representing predictions as Dirichlet distributions under subjective logic [31], EDL enables explicit estimation of both belief and uncertainty, providing a principled alternative to conventional probabilistic classifiers [32], [33]. Nevertheless, its application to HSCD remains largely unexplored, particularly in scenarios, where heterogeneous spectral–spatial representations exhibit strong inconsistency. Existing fusion strategies typically assume coherent feature responses and lack mechanisms to explicitly characterize and resolve such conflicts, which may undermine the reliability of final predictions [34], [35].

To overcome these challenges, we propose CCEM-Net, a conflict-aware EDL framework for reliable HSCD. Our framework is centered around a novel conflict-aware cross evidential modulation (CCEM) mechanism, which is designed to explicitly characterize and resolve conflicting evidence arising from heterogeneous spectral–spatial representations. Specifically, a change-aware interaction (CAI) module is first introduced to suppress pseudo-changes and enhance change-relevant responses through a dual-gating strategy. Next, a parallel adaptive multiscale spectral and spatial (AMSS) module captures diverse patterns by adaptively aggregating multiscale receptive fields guided by temporal differences. Finally, the CCEM mechanism models predictive outputs as Dirichlet evidence, explicitly measures evidential divergence via Jensen–Shannon (JS) divergence, and adaptively modulates evidence strength according to the degree of conflict, enabling conflict-aware reasoning for robust decision-making. Reliable decision-level fusion is then achieved using the Yager rule.

The main contributions of this work are summarized as follows.

1) We propose a novel CCEM-Net framework for HSCD, which introduces EDL to enable uncertainty-aware modeling and improve predictive reliability under bitemporal variations.
2) We develop a CCEM mechanism to characterize and resolve conflicting evidence in spectral–spatial representations, enabling adaptive evidence modulation and reliable decision-level fusion.
3) We design a CAI module and a parallel AMSS module to suppress pseudo-changes and capture discriminative patterns via difference-guided and multiscale representation learning.

Extensive experiments on benchmark HSCD datasets demonstrate that the proposed CCEM-Net consistently outperforms state-of-the-art (SOTA) methods in both detection accuracy and predictive reliability.

II. Related Works

A. Spectral–Spatial Representation and Bitemporal Modeling for HSCD

Recent advances in HSCD have been largely driven by deep learning, with a focus on spectral–spatial representation and bitemporal modeling. Quan et al. [36] highlighted the inherent nonuniformity of hyperspectral data across spectral, spatial, and feature dimensions, motivating adaptive representation learning.

To model bitemporal dependencies, difference-guided and interaction-based strategies have been widely explored. Wu et al. [18] proposed DIEFEN for differential feature aggregation and further developed AIWSEN [19] and CGMNet [20] to enhance interaction via entropy weighting and gated mechanisms. Chen et al. [37], [38] further explored temporal difference modeling and transformer-based architectures to capture multilevel differences and global spectral–spatial dependencies. Wang et al. [39] incorporated global and local attention, while Sun et al. [40] improved structural consistency through frequency-aware modeling. However, these methods generally assume consistent responses across temporal branches and lack mechanisms to handle discrepancies caused by spectral shifts and spatial heterogeneity.

To improve robustness, Chen et al. [21] disentangled temporal style variations in the frequency domain, while Feng et al. [22] enhanced subtle differences under noise. Zhang et al. [4] proposed a SAM-prior-guided spatial–spectral–temporal Mamba framework to improve robustness against affine misregistration in bitemporal HSIs. Nevertheless, these approaches remain deterministic and do not explicitly address conflicting evidence across spectral–spatial representations.

In parallel, multiscale feature extraction has become a prevalent strategy for capturing land-cover changes at diverse spatial scales. Representative architectures include hierarchical multiscale spectral–spatial aggregation [41], cross-guided multibranch feature fusion [42], and efficient spatio-temporal interaction modeling [43]. Although these methods effectively exploit complementary receptive fields, they generally employ a fixed multiscale aggregation strategy after training. The aggregation weights are learnable but are not explicitly conditioned on cross-temporal discrepancies, resulting in the same aggregation mechanism being applied regardless of local change characteristics.

The proposed AMSS differs in two design aspects. First, rather than employing fixed posttraining aggregation weights, AMSS conditions its gating coefficients on cross-temporal discrepancy features, allowing the receptive field to adapt to local change patterns. Second, AMSS performs this aggregation independently in spectral and spatial branches, preserving modality-specific characteristics before fusion. These design choices are tailored to the specific requirements of HSCD, where change patterns vary substantially in scale and spectral signature.

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Overall, although existing methods have achieved notable progress in spectral–spatial representation and bitemporal modeling, they generally assume coherent feature responses and lack the ability to quantify uncertainty or resolve conflicting evidence. Consequently, they may produce overconfident predictions in complex scenarios, motivating the development of uncertainty-aware and conflict-aware frameworks for reliable HSCD.

B. EDL and Uncertainty Modeling

Deterministic neural networks generally lack a principled and explicit mechanism to quantify predictive uncertainty [30]. To address this limitation, EDL was developed based on subjective logic, which models class probabilities using a Dirichlet distribution to explicitly represent both belief and uncertainty [30], [31]. To further improve this framework, Chen et al. [44] proposed Re-EDL, which alleviates overconfidence and suboptimal regularization by directly optimizing Dirichlet expectations. Compared with Bayesian approaches that rely on computationally expensive sampling [45], EDL provides a more efficient and practical solution for uncertainty estimation.

Beyond EDL, the importance of reliable confidence estimation has also been recognized in HSI analysis. Yang et al. [46] proposed CABIN to alleviate confirmation bias through confidence-aware semi-supervised learning, further highlighting the need for uncertainty-aware prediction.

Due to the high dimensionality, spectral redundancy, and complex variability across acquisition conditions, hyperspectral image (HSI) analysis inherently involves significant uncertainty, making uncertainty-aware modeling particularly important [47]. Recent studies have introduced EDL into HSI classification and unmixing tasks to better capture data variability [48]. For example, Feng et al. [49] proposed HyperEDL to address the epistemic uncertainty in cross-scene scenarios, while Qin et al. [50] developed an uncertainty-guided fusion strategy to suppress noise interference. However, these approaches primarily address intrafeature uncertainty in classification tasks and do not explicitly model predictive conflicts across heterogeneous feature branches.

To integrate multisource information, evidential fusion based on Dempster–Shafer (D–S) theory has been widely explored [51]. Xu et al. [52] proposed ECML, which learns view-specific evidence and aggregates subjective opinions through conflictive opinion aggregation (COA). COA theoretically guarantees that, for conflictive instances, the reliability of the aggregated opinion is lower than that of each individual view, thereby preventing confidence amplification. Nevertheless, the standard Dempster's combination rule may produce counter-intuitive results under high-conflict conditions [53]. To address this issue, Yager's rule transfers conflicting evidence into total ignorance, improving robustness [53], while Xiao [54] introduced JS divergence to measure evidence discrepancy for adaptive reliability weighting.

CCEM-Net differs from ECML in two aspects. First, ECML models view reliability through the uncertainty component of each subjective opinion, whereas CCEM explicitly models interbranch conflict via JS divergence, enabling different treatment of low-confidence agreement and high-confidence disagreement. Second, ECML directly aggregates subjective opinions, whereas CCEM first performs conflict-aware evidence discounting and subsequently applies Yager's rule, which transfers conflicting mass into uncertainty. This design may be particularly relevant to HSCD, where spectral and spatial branches can produce conflicting yet confident predictions around heterogeneous boundaries.

Overall, although EDL and evidential fusion provide principled uncertainty modeling, existing methods seldom consider explicit conflict modeling between heterogeneous branches, motivating the conflict-aware evidential framework proposed in this work.

III. Proposed Methodology

A. Overview

The overall architecture of the proposed CCEM-Net is illustrated in Fig. 1. Given a pair of co-registered bitemporal hyperspectral patches $\mathbf{X}_1, \mathbf{X}_2 \in \mathbb{R}^{C\times L\times L}$ acquired at times t_1 and t_2, where C is the number of spectral bands and $L \times L$ is the spatial size, our goal is to predict the change probability y of the central pixel.

The processing pipeline of CCEM-Net consists of three core stages. First, a CAI module explicitly models cross-temporal dependencies to suppress pseudo-changes while enhancing change-sensitive features. Second, the recalibrated features are processed by the AMSS module, which employs decoupled branches to capture heterogeneous spectral variations and spatial structural patterns via dynamic receptive-field aggregation. Finally, a CCEM mechanism quantifies predictive uncertainty and resolves interbranch conflicts using the Yager combination rule, yielding a reliable decision-level fusion.

B. CAI Module

Standard feature differencing is susceptible to illumination variations and registration errors. To mitigate these effects, the CAI module couples bitemporal observations through a dual-gating mechanism, as shown in Fig. 2.

1) Contextual Discrepancy Modeling: To inject local spatial context while maintaining spectral purity, we first apply a depthwise spatial mixing operator $\phi_{dw}(\cdot)$ to each temporal branch. A residual connection is employed to stabilize early training

$$\tilde{\mathbf{X}}_t = \mathbf{X}_t + \lambda_m \cdot \phi_{dw}(\mathbf{X}_t), \quad t \in \{1, 2\} \tag{1}$$

where λ_m is a learnable scaling factor. The contextualized temporal difference map is then computed as $\mathbf{D} = |\tilde{\mathbf{X}}_1 - \tilde{\mathbf{X}}_2|$.

2) Dual-Gating Interaction: The CAI module employs two parallel gates to recalibrate features from both spectral and spatial perspectives: spectral channel gate and spatial token gate.

To model cross-temporal spectral dependencies, we concatenate the bitemporal features with their discrepancy map. A positionwise MLP estimates the channelwise modulation coefficients

$$\left[\mathbf{G}_{c,1}, \mathbf{G}_{c,2}\right] = \sigma\left(\mathrm{MLP}\left(\left[\tilde{\mathbf{X}}_1, \tilde{\mathbf{X}}_2, \mathbf{D}\right]\right)\right) \in \mathbb{R}^{2C\times L\times L} \tag{2}$$

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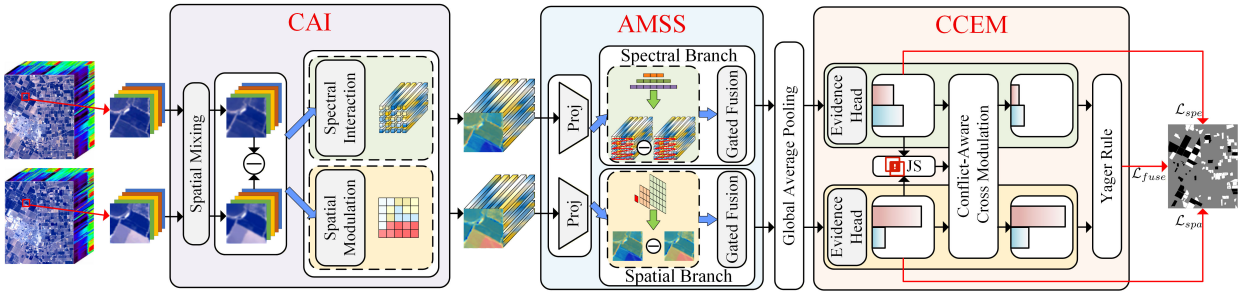


Fig. 1. Overall architecture of the proposed CCEM-Net, which sequentially performs cross-temporal interaction via the CAI module, dual-stream multiscale representation learning through the AMSS module, and conflict-aware decision fusion using the CCEM mechanism.

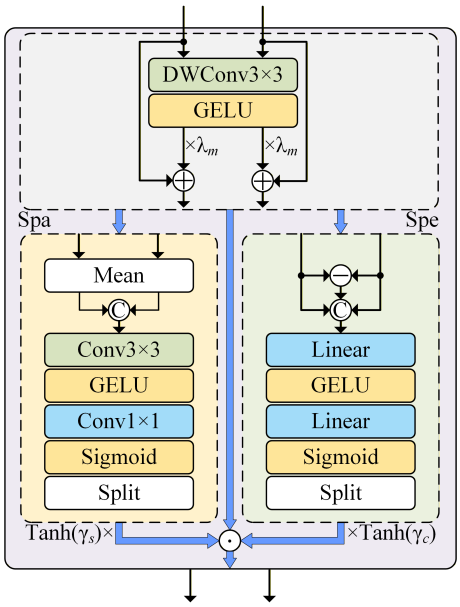


Fig. 2. Detailed structure of the CAI module with dual-gating mechanism.

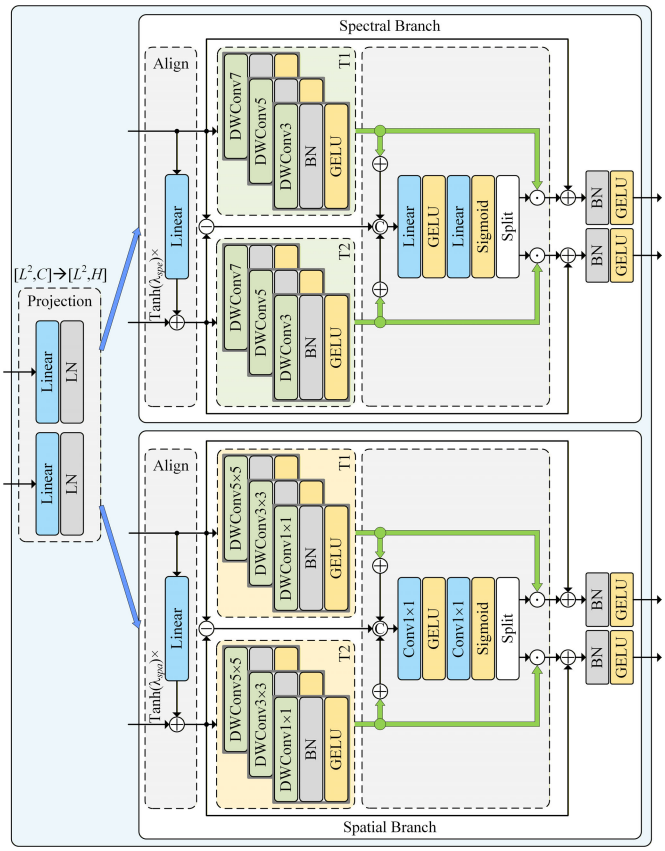


Fig. 3. Structural components of the AMSS module.

where $\sigma(\cdot)$ is the sigmoid function. This gate adaptively amplifies discriminative bands that exhibit significant and consistent change signatures across time.

Simultaneously, a spatial gate is introduced to emphasize spatially coherent changes. We first compress the spectral dimension via average pooling to obtain compact spatial descriptors $\mathbf{T} = [\mathrm{Avg}_c(\tilde{\mathbf{X}}_1), \mathrm{Avg}_c(\tilde{\mathbf{X}}_2)] \in \mathbb{R}^{2\times L\times L}$. A convolutional block $\phi_s(\cdot)$ then generates spatial modulation maps

$$\left[\mathbf{G}_{s,1}, \mathbf{G}_{s,2}\right] = \sigma\left(\phi_s\left(\mathbf{T}\right)\right) \in \mathbb{R}^{2\times L\times L}. \tag{3}$$

3) Feature Recalibration: The final interaction-aware features are obtained by modulating the contextualized inputs with the estimated spectral and spatial gates. To ensure the optimization stability, we employ learnable scaling parameters γ_c and γ_s within a residual framework

$$\mathbf{X}'_t = \tilde{\mathbf{X}}_t \odot \left(1 + \tanh\left(\gamma_c\right)\mathbf{G}_{c,t}\right) \odot \left(1 + \tanh\left(\gamma_s\right)\mathbf{G}_{s,t}\right) \tag{4}$$

where $t \in \{1, 2\}$ and $\odot$ denotes the elementwise multiplication. By initializing the scaling factors γ_c and γ_s at 0, the CAI module effectively starts as an identity mapping and gradually learns to suppress pseudo-changes while amplifying change-sensitive responses.

C. AMSS Module

To address the heterogeneous spatial scales and complex spectral correlations of land-cover changes, we develop an AMSS module. As illustrated in Fig. 3, AMSS employs a decoupled architecture consisting of parallel spectral and spatial branches, each performing change-aware multiscale refinement. Rather than employing fixed posttraining aggregation weights, AMSS conditions its multiscale gating coefficients on cross-temporal discrepancy features. As a result, the effective receptive field is adaptively adjusted according to local change characteristics, allowing the model to capture both subtle and large-scale land-cover changes.

1) Representation Embedding and Cross-Temporal Alignment: The input features are first projected into a compact

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embedding space to enhance discriminative semantics

$$\mathbf{F}_t = \mathrm{LN}\left(\mathbf{X}_t'^{\top}\mathbf{W}_r\right), \quad \mathbf{F}_t \in \mathbb{R}^{L^2\times H} \tag{5}$$

where $\mathbf{W}_r \in C \times H$ is the projection matrix and LN$(\cdot)$ denotes layer normalization. To mitigate residual cross-temporal appearance discrepancies caused by imaging conditions, we introduce a tanh-gated alignment mechanism

$$\hat{\mathbf{F}}_{b,2} = \mathbf{F}_{b,2} + \tanh(\lambda_b)\ \phi_b\left(\mathbf{F}_{b,1}\right), \quad b \in \{spe, spa\} \tag{6}$$

where $\phi_b(\cdot)$ is a learnable mapping and λ_b is an alignment-strength parameter. The bounded gate $\tanh(\lambda_b) \in (-1, 1)$ ensures identity initialization, gradually enabling alignment during training to preserve optimization stability.

2) Decoupled Multiscale Feature Extraction: The AMSS module extracts multiscale representations through parallel depthwise convolutional streams, where the spectral branch employs kernel sizes $\mathcal{K}_{spe} = \{3, 5, 7\}$ and the spatial branch uses $\mathcal{K}_{spa} = \{1, 3, 5\}$.

To capture long-range spectral dependencies and suppress localized noise, 1-D depthwise convolutions are applied along the embedding dimension

$$\mathbf{F}_1^{(k)} = \phi_{spe}^{(k)}\left(\mathbf{F}_1\right), \quad \mathbf{F}_2^{(k)} = \phi_{spe}^{(k)}\left(\hat{\mathbf{F}}_{spe,2}\right), \quad k \in \mathcal{K}_{spe}. \tag{7}$$

To preserve fine-grained edge details and fragmented structural transitions, 2-D depthwise convolutions are applied to the reshaped spatial lattice $\mathbf{U}_1$ and aligned $\hat{\mathbf{U}}_2 \in \mathbb{R}^{H\times L\times L}$

$$\mathbf{S}_1^{(k)} = \phi_{spa}^{(k)}\left(\mathbf{U}_1\right), \quad \mathbf{S}_2^{(k)} = \phi_{spa}^{(k)}\left(\hat{\mathbf{U}}_2\right), \quad k \in \mathcal{K}_{spa}. \tag{8}$$

Standard zero padding is applied to preserve spatial resolution. When $L = 1$, the spatial depthwise convolutions reduce to per-channel scaling, as only the center kernel weight contributes under zero padding. Although the spatial context is absent, kernels of different sizes still yield distinct per-channel modulation coefficients, and the dynamic gating in (10) selectively weights them.

3) Change-Aware Dynamic Gating: Rather than using fixed receptive fields, AMSS adaptively aggregates multiscale features guided by temporal difference cues. We first compute the multiscale summaries $\mathbf{M}_t = \sum_{k\in\mathcal{K}} \mathbf{F}_t^{(k)}$ for each branch. A change-aware gating mechanism then predicts positionwise coefficients based on the interaction of bitemporal summaries and their absolute discrepancy

$$\mathbf{G}_b = \sigma\left(\mathrm{MLP}\left(\left[\mathbf{M}_{b,1}, \mathbf{M}_{b,2}, |\mathbf{F}_{b,1} - \hat{\mathbf{F}}_{b,2}|\right]\right)\right) \tag{9}$$

where $[\cdot]$ denotes the concatenation. The six-channel coefficients in $\mathbf{G}_b$ are split into two sets $\{g_t^{(k)}\}$ for the respective temporal streams. The final refined features for branch b are computed as a weighted residual sum

$$\mathbf{F}_t^b = \mathrm{BN}\left(\sum_{k\in\mathcal{K}_b} g_t^{(k)} \odot \mathbf{F}_t^{(k)} + \mathbf{F}_t\right) \tag{10}$$

where BN$(\cdot)$ denotes batch normalization. This dynamic aggregation enables the network to adaptively adjust its receptive configuration based on the local magnitude and scale of changes, ensuring the discriminative spectral–spatial representation for subsequent fusion.

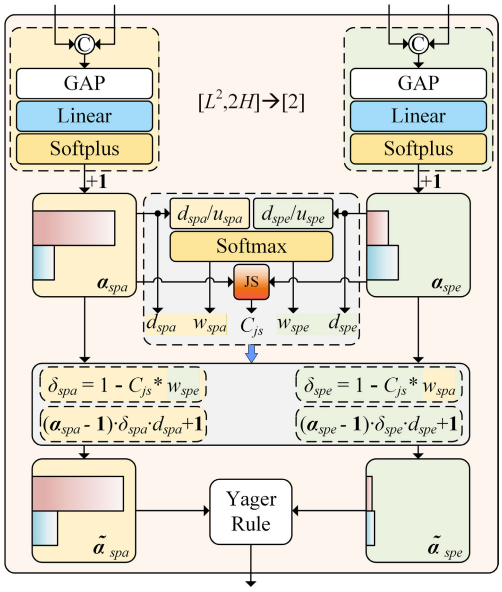


Fig. 4. Framework of the CCEM mechanism for evidence fusion.

D. Conflict-Aware Cross Evidential Modulation

To improve decision reliability in ambiguous regions, the representations from the spectral and spatial branches are not fused directly. Instead, they are modeled as independent Dirichlet distributions. We propose a CCEM mechanism, as illustrated in Fig. 4, that explicitly quantifies predictive uncertainty, estimates interbranch conflict, and dynamically modulates evidence prior to fusion via the Yager combination rule.

1) Evidence Modeling and Uncertainty Estimation: First, global descriptors are extracted from the refined spectral and spatial features via global average pooling (GAP) over the spatial dimensions

$$\mathbf{z}_{spe} = \mathrm{Avg}\left(\mathbf{F}_{spe}\right), \quad \mathbf{z}_{spa} = \mathrm{Avg}\left(\mathbf{F}_{spa}\right) \in \mathbb{R}^{N\times 2H}. \tag{11}$$

Two independent linear evidential heads map these descriptors to branch-specific evidence logits. Following EDL principles, nonnegative evidence vectors are obtained using the Softplus activation function

$$\mathbf{e}_b = \mathrm{Softplus}\left(\mathbf{W}_b\mathbf{z}_b\right), \quad b \in \{spe, spa\} \tag{12}$$

where $\mathbf{W}_b \in \mathbb{R}^{K\times 2H}$ is the classifier weight for branch b, and K denotes the number of classes. The parameters of the corresponding Dirichlet distribution are subsequently defined as $\boldsymbol{\alpha}_b = \mathbf{e}_b + \mathbf{1}$.

Let $S_b = \sum_{k=1}^{K} \alpha_{b,k}$ be the total Dirichlet strength. The predictive uncertainty of branch b is mathematically formulated as follows:

$$u_b = \frac{K}{S_b}. \tag{13}$$

This formulation reflects the overall confidence of the evidence: a larger S_b indicates stronger accumulated evidence and thus lower uncertainty, while a smaller S_b corresponds to insufficient evidence and higher predictive uncertainty.

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Concurrently, the expected categorical probability for the kth class is given by $p_{b,k} = \alpha_{b,k}/S_b$. To evaluate the determinacy of the prediction, we compute the entropy of the expected probability distribution, denoted by $\mathcal{H}_b = -\sum_{k=1}^{K} p_{b,k} \log p_{b,k}$. The determinacy score is then defined as follows:

$$d_b = \frac{1}{\mathcal{H}_b + \varepsilon} \tag{14}$$

where ε is a small constant for numerical stability. This score measures the concentration of the predicted distribution: lower entropy leads to higher determinacy, indicating more decisive predictions, whereas higher entropy results in lower determinacy, reflecting ambiguous or uncertain class assignments.

2) Conflict-Aware Cross Modulation: Although the dual branches provide complementary cues, they may yield contradictory predictions in complex scenarios, such as boundary regions. To quantify this cognitive disagreement, we compute the normalized JS divergence between their expected distributions

$$C_{js} = \frac{1}{\log K}\left(\frac{1}{2}D_{KL}\left(\mathbf{p}_{spa}\|\mathbf{m}\right) + \frac{1}{2}D_{KL}\left(\mathbf{p}_{spe}\|\mathbf{m}\right)\right) \tag{15}$$

where $\mathbf{m} = 1/2(\mathbf{p}_{spa} + \mathbf{p}_{spe})$. The normalized JS divergence ensures a symmetric and bounded conflict measurement $C_{js} \in [0, 1]$.

Rather than penalizing both branches equally, CCEM adaptively modulates the evidence based on the relative branch reliability. We define the relative confidence score as $v_b = d_b/u_b$, which simultaneously encourages high determinacy and low uncertainty. The branch-specific scaling weights are generated via a temperature-scaled softmax function

$$w_b = \frac{\exp\left(v_b/\tau\right)}{\sum_{b\in\{spe,spa\}} \exp\left(v_b/\tau\right)} \tag{16}$$

where τ is a learnable temperature parameter. A cross-evidential discount factor is then formulated to suppress the evidence of the unreliable branch when conflict occurs

$$\delta_{spe} = \max\left(1 - C_{js} \cdot w_{spa},\, \ell\right) \tag{17}$$

$$\delta_{spa} = \max\left(1 - C_{js} \cdot w_{spe},\, \ell\right) \tag{18}$$

where $\ell = 0.01$ serves as a lower bound constraint. This mechanism ensures that if high conflict exists and one branch (e.g., spatial) is highly confident, the evidence of the opposing branch (e.g., spectral) is heavily discounted. The modulated evidence for fusion is explicitly adjusted by both the discount factor and the determinacy score

$$\tilde{\boldsymbol{\alpha}}_b = (\boldsymbol{\alpha}_b - \mathbf{1}) \cdot \delta_b \cdot d_b + \mathbf{1}. \tag{19}$$

3) Decision Fusion via Yager Rule: To effectively fuse the modulated evidence while preserving uncertainty semantics, we employ the Yager combination rule. Unlike standard Dempster's rule, the Yager rule assigns conflicting mass to the universal set (uncertainty) rather than redistributing it among focal elements, thereby avoiding overly overconfident erroneous predictions.

Let the basic probability assignment (BPA) and the uncertainty mass of the modulated branch b be $\tilde{\mathbf{b}}_b = (\tilde{\boldsymbol{\alpha}}_b - \mathbf{1})/\tilde{S}_b$ and $\tilde{u}_b = K/\tilde{S}_b$, respectively. The fused belief mass $\mathbf{b}_{fuse}$ is computed as follows:

$$\mathbf{b}_{fuse} = \tilde{\mathbf{b}}_{spe} \odot \tilde{\mathbf{b}}_{spa} + \tilde{\mathbf{b}}_{spe}\tilde{u}_{spa} + \tilde{\mathbf{b}}_{spa}\tilde{u}_{spe} \tag{20}$$

where $\odot$ denotes the elementwise multiplication. The degree of conflict c between the two modulated branches is aggregated to the fused uncertainty mass u_{fuse}

$$u_{fuse} = \tilde{u}_{spe}\tilde{u}_{spa} + \sum_{i=1}^{K}\sum_{j\neq i} \tilde{b}_{spe,i}\tilde{b}_{spa,j}. \tag{21}$$

The final fused Dirichlet parameters are subsequently reconstructed as follows:

$$\boldsymbol{\alpha}_{fuse} = \mathbf{b}_{fuse}\frac{K}{u_{fuse}} + \mathbf{1}. \tag{22}$$

During inference, rather than directly outputting expected probabilities, the network generates the expected log-probability for the final classification

$$\hat{y}_k = \mathbb{E}\left[\log p_k\right] = \psi\left(\alpha_{fuse,k}\right) - \psi\left(\sum_{j=1}^{K}\alpha_{fuse,j}\right) \tag{23}$$

where $\psi(\cdot)$ is the digamma function. This formulation yields a numerically stable decision function while rigorously maintaining the modeled predictive uncertainty.

E. Training Objective

The proposed network is trained end-to-end using a joint evidential objective, which consists of a primary fused evidential loss and branch-level auxiliary supervision.

Let $\mathbf{y} \in \{0, 1\}^K$ be the one-hot encoded ground-truth label for a given pixel, and let $\boldsymbol{\alpha}_{fuse} \in \mathbb{R}^K$ be the fused Dirichlet parameters produced by the conflict-aware fusion module. According to the EDL framework, the primary data-fitting term is formulated as an evidential mse risk, defined as follows:

$$\mathcal{L}_{mse}\left(\boldsymbol{\alpha}_{fuse}, \mathbf{y}\right) = \sum_{k=1}^{K}\left[\left(y_k - \frac{\alpha_{fuse,k}}{S_{fuse}}\right)^2 + \frac{\alpha_{fuse,k}\left(S_{fuse} - \alpha_{fuse,k}\right)}{S_{fuse}^2\left(S_{fuse} + 1\right)}\right] \tag{24}$$

where $S_{fuse} = \sum_{k=1}^{K}\alpha_{fuse,k}$ represents the total evidence strength. The first term penalizes the prediction error, while the second term captures the variance of the Dirichlet distribution to enforce reliable uncertainty estimates.

To discourage misleading evidence assigned to nontarget classes, a Kullback–Leibler (KL) divergence regularization term is incorporated

$$\mathcal{L}_{kl}\left(\boldsymbol{\alpha}_{fuse}, \mathbf{y}\right) = D_{KL}\left(\mathrm{Dir}\left(\tilde{\boldsymbol{\alpha}}_{fuse}\right) \,\middle\|\, \mathrm{Dir}\left(\mathbf{1}\right)\right) \tag{25}$$

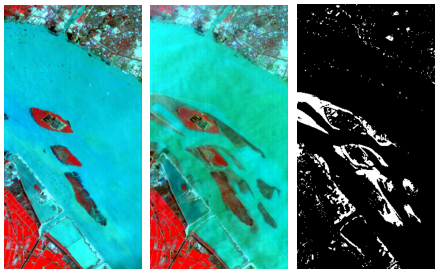
where $\tilde{\boldsymbol{\alpha}}_{fuse} = \mathbf{y} + (1 - \mathbf{y}) \odot \boldsymbol{\alpha}_{fuse}$ explicitly masks the evidence of the true class. This ensures that only the false evidence is penalized against a uniform Dirichlet prior Dir($\mathbf{1}$). The primary fused evidential objective is thus computed as follows:

$$\mathcal{L}_{EDL}\left(\boldsymbol{\alpha}, \mathbf{y}\right) = \mathcal{L}_{mse}\left(\boldsymbol{\alpha}, \mathbf{y}\right) + \eta_t\mathcal{L}_{kl}\left(\boldsymbol{\alpha}, \mathbf{y}\right) \tag{26}$$

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TABLE I
DATASET STATISTICS AND LEGEND

	River [16]	Farmland [55]	Bay Area [56]
Size	463 × 241	450 × 140	600 × 500
Bands	198	155	224
Changed	9698	18277	39270
Unchanged	101885	44723	34211
Unknown	0	0	226519



(a) (b) (c)

Fig. 5. Visualization of the River dataset. (a) Image at t_1. (b) Image at t_2. (c) Ground-truth map.

where η_t is an annealing coefficient defined as follows:

$$\eta_t = \min\left(1, \frac{t}{0.5T}\right) \tag{27}$$

where t and T denote the current and total training epochs, respectively. Thus, η_t linearly increases from 0 to 1 during the first half of training and remains 1 afterward, preventing overly strong KL regularization at early epochs.

Furthermore, to prevent branch collapse and ensure that the spectral and spatial branches maintain discriminative standalone representations, we impose auxiliary evidential supervision

$$\mathcal{L}_{aux} = \mathcal{L}_{EDL}\left(\boldsymbol{\alpha}_{spe}, \mathbf{y}\right) + \mathcal{L}_{EDL}\left(\boldsymbol{\alpha}_{spa}, \mathbf{y}\right). \tag{28}$$

The overall training objective of CCEM-Net is defined as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{EDL}\left(\boldsymbol{\alpha}_{fuse}, \mathbf{y}\right) + \lambda_{aux}\mathcal{L}_{aux} \tag{29}$$

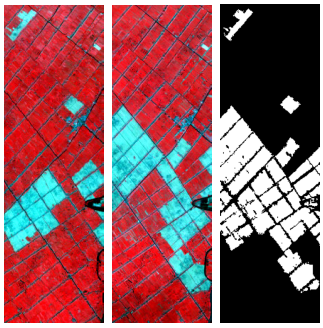
where λ_{aux} is a hyperparameter balancing the evidence fusion optimization and unimodal representation learning. This joint objective effectively mitigates branch degradation and facilitates conflict-aware uncertainty modeling.

IV. EXPERIMENTS

A. Dataset Description

To evaluate the proposed method, we conduct experiments on three widely used bitemporal hyperspectral datasets—River [16], Farmland [55], and Bay Area [56]—with details summarized in Table I.

1) River: This dataset was collected by the EO-1 Hyperion sensor in 2013 over a riverine area in Jiangsu, China. It has a spatial size of 463 × 241 pixels with 198 spectral bands after noise removal. The primary change type is riverbank recession. The visual representation and GT map are provided in Fig. 5.



(a) (b) (c)

Fig. 6. Visualization of the Farmland dataset. (a) Image at t_1. (b) Image at t_2. (c) Ground-truth map.



(a) (b) (c)

Fig. 7. Visualization of the Bay Area dataset. (a) Image at t_1. (b) Image at t_2. (c) Ground-truth map.

2) Farmland: This dataset was acquired by the EO-1 Hyperion sensor in 2006 and 2007 over an agricultural region near Yancheng, Jiangsu, China. It features a spatial size of 450 × 140 pixels with 155 spectral bands after preprocessing. The main changes correspond to variations in cultivation areas and crop types. The visual representation and GT map are provided in Fig. 6.

3) Bay Area: This dataset was collected via the AVIRIS sensor in 2013 and 2015 over Patterson, California. It features a spatial resolution of 600 × 500 pixels with 224 spectral bands. The main change categories involve urban development and farmland transitions. The visual representation and GT map are provided in Fig. 7.

B. Experimental Setup

1) Implementation Details: To ensure a fair comparison, the CCEM-Net and all benchmark methods were implemented via PyTorch 2.6.0 on a server with an Intel Xeon E5-2680 v4 CPU and an NVIDIA RTX 3090 GPU. For each dataset, we randomly sampled only 1% of labeled pixels for training, leaving the remaining 99% for testing to evaluate the model generalization under limited supervision. Training was performed using the Adam optimizer for 200 epochs with a batch size of 64. The initial learning rate was 1×10^{-4}, decayed to 1×10^{-6} using a cosine annealing schedule. Model parameters yielding the minimum training loss were preserved for final assessment.

2) Evaluation Indicators: The performance is assessed from two complementary perspectives: classification accuracy and probabilistic reliability. For categorical correctness, we employ overall accuracy (OA), the Kappa coefficient (κ), and

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the F1-score (F1), where κ and F1 provide robust evaluations under class imbalance.

To evaluate uncertainty quantification, we adopt expected calibration error (ECE), negative log-likelihood (NLL), and the area under the risk-coverage curve (RC-AUC).

Specifically, ECE measures the discrepancy between predictive confidence and empirical accuracy by partitioning predictions into M bins, where $M = 20$ in this work

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} |\mathrm{acc}(B_m) - \mathrm{conf}(B_m)| \tag{30}$$

where the confidence $\mathrm{conf}(B_m)$ is obtained from the Dirichlet expectation $\mathbb{E}[p_k] = \alpha_k / S$. This metric evaluates how well the evidential predictions are calibrated.

NLL evaluates the probabilistic consistency of the model predictions. Given that the network directly outputs the expected log-probability as in (23), NLL is computed as follows:

$$\mathrm{NLL} = -\frac{1}{N} \sum_{i=1}^{N} \left[\psi\left(\alpha_{y_i}\right) - \psi(S) \right] \tag{31}$$

which naturally aligns with the evidential formulation and provides a measure of how overconfident incorrect predictions affect the probabilistic quality.

Furthermore, RC-AUC assesses the model's ability to rank samples according to predictive uncertainty. Using the confidence derived from Dirichlet expectation, samples are sorted in descending order, and the coverage and risk at position k are defined as follows:

$$\mathrm{Coverage}(k) = \frac{k}{N}, \quad \mathrm{Risk}(k) = 1 - \frac{1}{k} \sum_{i=1}^{k} \mathbb{I}(\hat{y}_i = y_i). \tag{32}$$

The RC-AUC is then computed as follows:

$$\text{RC-AUC} = \int_0^1 \mathrm{Risk}(c)\, DC \tag{33}$$

where a lower value indicates that the model can effectively assign higher uncertainty to erroneous predictions.

Together, these metrics provide a comprehensive evaluation of both predictive accuracy and the quality of uncertainty estimation, which is essential for assessing the reliability of evidential models in HSCD.

C. Hyperparameter Analysis

To evaluate the sensitivity of CCEM-Net to key hyperparameters, we analyze three factors: spatial patch size L, hidden dimension H, and auxiliary loss coefficient λ_{aux}. The effects of L and H are assessed by OA, while λ_{aux} is evaluated using both OA and ECE to reflect accuracy–reliability tradeoffs.

1) Impacts of Patch Size: The patch size $L \in \{1, 3, 5, 7, 9\}$ controls spatial context. As shown in Fig. 8(a), the optimal L varies across datasets. River achieves the best performance at $L = 3$, indicating a preference for localized context. Farmland peaks at $L = 1$ and degrades with larger patches; at $L = 1$, the spatial depthwise convolutions degenerate to per-channel scaling, suggesting that farmland changes are characterized by subtle spectral variations that are easily diluted by spatial context. In contrast, Bay Area benefits significantly from larger context, reaching optimal performance at $L = 9$. This divergence highlights the dataset-dependent preference for spectral and spatial information, thereby justifying the necessity of the decoupled spectral–spatial modeling in AMSS.

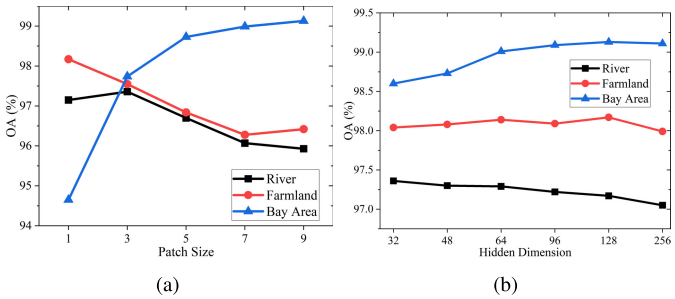


Fig. 8. Sensitivity analysis of hyperparameters. (a) Impact of patch size L on OA. (b) Impact of hidden dimension H on OA.

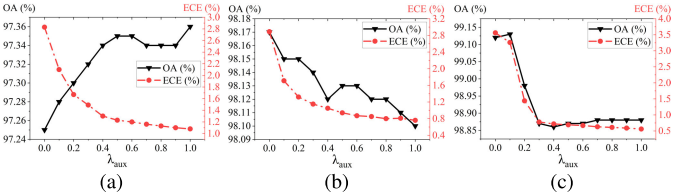


Fig. 9. Sensitivity analysis of the auxiliary loss coefficient λ_{aux} across three datasets. (a) River. (b) Farmland. (c) Bay Area.

2) Impact of Hidden Dimension: The hidden dimension $H \in \{32, 64, 96, 128, 256\}$ controls representation capacity. As illustrated in Fig. 8(b), CCEM-Net remains highly stable across different H. River achieves optimal performance at $H = 32$, while Farmland and Bay Area peak at $H = 128$. Notably, performance variations remain within 0.5%, demonstrating that the model is not sensitive to embedding width and can consistently learn discriminative evidential representations.

3) Influence of Auxiliary Loss Coefficient λ_{aux}: The coefficient λ_{aux} balances joint fusion and branch-level supervision. As shown in Fig. 9, increasing λ_{aux} significantly reduces ECE across all datasets, indicating improved reliability and reduced overconfidence.

In terms of OA, River benefits from stronger supervision, while Farmland and Bay Area exhibit slight performance drops at larger λ_{aux}. This is attributed to modality dominance, where enforcing strong supervision on weaker branches introduces mild regularization overhead. Nevertheless, the marginal OA reduction is outweighed by substantial gains in reliability, making λ_{aux} an effective tradeoff controller.

4) Analysis of the Learnable Temperature τ: The learnable temperature τ controls the sharpness of branch weighting in the proposed CCEM mechanism. It is initialized to 1.0 and jointly optimized with the network parameters throughout training. To prevent excessively sharp or overly smooth weighting distributions, τ is constrained to the range [0.01, 10] during optimization.

To evaluate the robustness of the learnable temperature, we initialize τ with different values while keeping all other

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TABLE II
SENSITIVITY TO THE INITIALIZATION OF THE LEARNABLE TEMPERATURE τ

Initial τ	River	Farmland	Bay Area
	OA / ECE (%)	OA / ECE (%)	OA / ECE (%)
0.2	97.35 / 1.67	98.10 / 0.78	98.87 / 1.06
0.5	97.34 / 1.67	98.10 / 0.77	98.88 / 1.07
1.0	97.36 / 1.68	98.10 / 0.76	98.88 / 1.04
2.0	97.34 / 1.67	98.10 / 0.77	98.89 / 1.07
5.0	97.32 / 1.72	98.09 / 0.77	98.88 / 1.07

settings unchanged. As reported in Table II, both OA and ECE remain nearly unchanged across all three datasets, even when the initialization varies from 0.2 to 5.0. These results indicate that the proposed CCEM-Net is insensitive to the initialization of τ.

Under the default initialization ($\tau = 1.0$), the learned temperatures are 1.0605 ± 0.0270, 0.9801 ± 0.0049, and 0.9631 ± 0.0016 on the River, Farmland, and Bay Area datasets, respectively. The small standard deviations demonstrate stable optimization and consistent convergence across different runs.

Considering both accuracy and reliability, we determine the final hyperparameter configurations for all subsequent experiments. Specifically, for the River, Farmland, and Bay Area datasets, the patch size L is set to 3, 1, and 9, respectively, while the hidden dimension H is configured as 32 for River and 128 for both Farmland and Bay Area. The auxiliary loss weight λ_{aux} is uniformly fixed at 1.0 across all datasets. The learnable temperature τ is initialized to 1.0 in all experiments.

D. Comparison With SOTA

To validate the effectiveness of CCEM-Net, we compare it with nine SOTA HSCD methods, include the following.

1) MFCEN [15], a multilevel feature fusion and change information enhancement network.
2) FDIEG [21], a frequency-domain information exchange and generation network with a Siamese U-shaped structure.
3) DIEFEN [18], a differential information-enhanced feature exchange network.
4) GLAFormer [39], a global and local attention-based Transformer.
5) AIWSEN [19], an adaptive information weighting and synchronized enhancement network.
6) BTSCD [57], a semantic change detection network based on boundary detection and task interaction.
7) WCEDNet [17], a weighted cascaded encoder–decoder network.
8) SFIEET [40], a spectral frequency-induced edge enhancement Transformer.
9) CGMNet [20], a center-pixel and gated mechanism-based attention network.

Notably, these existing approaches are all grounded in conventional softmax-based probabilistic frameworks and lack explicit uncertainty modeling. For a fair comparison, all methods are implemented using their official source codes and maintained with the recommended parameter settings.

1) Quantitative Comparison: Table III reports the quantitative results on three datasets. As shown in the table, CCEM-Net consistently achieves the best performance across all datasets and metrics. In terms of classification accuracy, it attains the highest OA, κ, and F1 scores. For example, on Bay Area, CCEM-Net reaches an OA of 98.88%, outperforming all competing methods. Similar gains are observed on River and Farmland, demonstrating the effectiveness of the proposed representation learning modules.

However, accuracy alone does not reflect decision reliability. As shown in Table III, several softmax-based methods (e.g., DIEFEN and AIWSEN) exhibit extremely high ECE values (over 20%), indicating severe overconfidence despite competitive accuracy. This reveals a fundamental limitation of conventional probabilistic frameworks.

In contrast, CCEM-Net models predictions as Dirichlet distributions, enabling the principled uncertainty quantification and achieving significantly improved calibration (e.g., ECE $\approx$ 1% across datasets). Building upon this, the proposed CCEM further enhances decision reliability by explicitly measuring interbranch inconsistency and adaptively adjusting evidence before fusion, thereby effectively mitigating overconfident predictions.

Overall, these results demonstrate that reliable HSCD requires not only high accuracy but also proper uncertainty modeling and conflict-aware reasoning, which are jointly achieved by CCEM-Net.

2) Qualitative Comparison: To visually demonstrate the effectiveness of the proposed framework, the change detection error maps of different methods on the three datasets are shown in Figs. 10–12. In these maps, white and black pixels denote true positives and true negatives, while red and blue pixels represent false positives and false negatives, respectively.

On the River dataset (see Fig. 10), complex and winding boundaries introduce high uncertainty and inconsistencies between spectral and spatial features. Consequently, several baseline methods, such as FDIEG, BTSCD, and SFIEET, exhibit noticeable missed detections and false alarms along boundaries. In contrast, CCEM-Net produces smoother and more continuous contours, effectively reducing boundary-related errors.

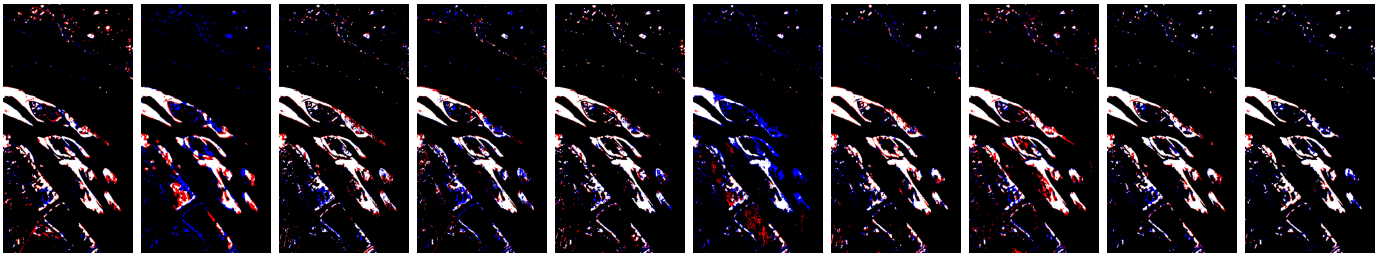
On the Farmland dataset (see Fig. 11), which is dominated by large homogeneous regions, methods such as FDIEG, DIEFEN, and BTSCD tend to produce fragmented structures with internal noise, while AIWSEN and SFIEET introduce false alarms near boundaries. In comparison, CCEM-Net maintains strong intraclass consistency, preserving structural integrity while suppressing spurious responses.

The advantage of CCEM-Net becomes more evident on the highly heterogeneous Bay Area dataset (see Fig. 12). Baseline methods such as DIEFEN and SFIEET generate numerous false alarms in unchanged regions, reflecting sensitivity to pseudo-changes. In contrast, CCEM-Net effectively suppresses these unreliable predictions and produces more precise and coherent change regions.

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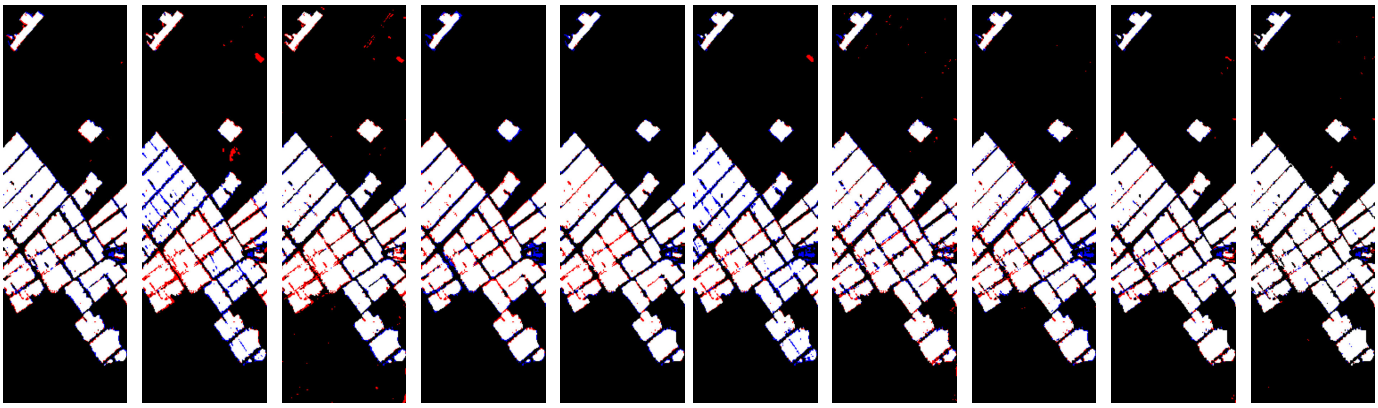
TABLE III
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THREE DATASETS

Dataset	Metric	MFCEN	FDIEG	DIEFEN	GLAFormer	AIWSEN	BTSCD	WCEDNet	SFIEET	CGMNet	CCEM-Net
River	OA (%) ↑	96.11 ± 0.12	93.72 ± 0.56	96.33 ± 0.24	95.92 ± 0.27	96.69 ± 0.15	95.27 ± 0.60	96.07 ± 0.39	95.07 ± 0.67	96.68 ± 0.26	$\mathbf{97.36 \pm 0.05}$
	$\kappa \times 100$ ↑	76.16 ± 0.56	56.08 ± 5.84	77.23 ± 0.83	73.12 ± 1.98	78.91 ± 0.74	67.50 ± 3.63	73.99 ± 5.37	70.79 ± 2.62	78.55 ± 2.13	$\mathbf{82.71 \pm 0.40}$
	F1 (%) ↑	88.08 ± 0.28	78.01 ± 2.95	88.61 ± 0.42	86.55 ± 0.99	89.45 ± 0.37	83.73 ± 1.82	86.97 ± 2.72	85.38 ± 1.32	89.28 ± 1.07	$\mathbf{91.35 \pm 0.20}$
	ECE (%) ↓	3.05 ± 0.16	2.06 ± 0.38	24.90 ± 0.29	3.54 ± 0.26	23.91 ± 0.19	3.84 ± 0.47	3.36 ± 0.27	4.25 ± 0.66	2.11 ± 0.39	$\mathbf{1.64 \pm 0.25}$
	NLL×100 ↓	15.55 ± 0.77	16.72 ± 0.92	35.73 ± 0.30	33.88 ± 2.66	34.61 ± 0.18	22.23 ± 4.20	29.62 ± 1.71	34.92 ± 7.73	13.32 ± 2.91	$\mathbf{8.70 \pm 0.41}$
	RC-AUC×10^3 ↓	4.41 ± 1.12	10.05 ± 0.85	4.16 ± 0.49	8.52 ± 1.30	19.82 ± 6.61	13.06 ± 7.52	6.38 ± 0.77	7.16 ± 1.41	5.18 ± 0.92	$\mathbf{1.78 \pm 0.12}$
Farm.	OA (%) ↑	96.77 ± 0.17	94.96 ± 2.22	96.16 ± 0.56	95.70 ± 0.57	97.40 ± 0.27	95.56 ± 0.86	96.42 ± 0.40	96.41 ± 0.24	97.27 ± 0.08	$\mathbf{98.10 \pm 0.08}$
	$\kappa \times 100$ ↑	92.18 ± 0.44	87.84 ± 5.99	90.78 ± 1.40	89.56 ± 1.37	93.73 ± 0.64	89.10 ± 2.15	91.33 ± 0.97	91.30 ± 0.59	93.41 ± 0.19	$\mathbf{95.40 \pm 0.19}$
	F1 (%) ↑	96.09 ± 0.22	93.92 ± 3.02	95.39 ± 0.70	94.78 ± 0.69	96.86 ± 0.32	94.55 ± 1.07	95.67 ± 0.48	95.65 ± 0.30	96.71 ± 0.10	$\mathbf{97.70 \pm 0.10}$
	ECE (%) ↓	2.80 ± 0.22	4.98 ± 4.38	25.24 ± 0.80	3.27 ± 0.64	24.63 ± 0.25	3.52 ± 1.01	3.10 ± 0.36	3.08 ± 0.24	1.58 ± 0.14	$\mathbf{0.76 \pm 0.09}$
	NLL×100 ↓	16.07 ± 1.84	26.90 ± 5.05	36.86 ± 0.75	26.65 ± 8.85	33.96 ± 0.32	19.90 ± 6.60	30.22 ± 8.62	23.97 ± 2.56	9.14 ± 0.31	$\mathbf{6.27 \pm 0.27}$
	RC-AUC×10^3 ↓	4.03 ± 0.83	32.87 ± 15.94	13.01 ± 4.69	7.09 ± 3.11	17.07 ± 6.61	10.93 ± 5.43	8.51 ± 8.83	3.61 ± 0.28	4.71 ± 1.75	$\mathbf{1.27 \pm 0.17}$
BayA.	OA (%) ↑	98.17 ± 0.25	95.19 ± 1.82	95.70 ± 0.37	96.85 ± 0.25	96.35 ± 0.27	97.20 ± 0.44	97.39 ± 1.95	96.02 ± 1.14	98.22 ± 0.14	$\mathbf{98.88 \pm 0.07}$
	$\kappa \times 100$ ↑	96.32 ± 0.51	90.33 ± 3.66	91.31 ± 0.74	93.67 ± 0.50	92.67 ± 0.54	94.37 ± 0.89	94.76 ± 3.91	91.97 ± 2.31	96.43 ± 0.29	$\mathbf{97.75 \pm 0.15}$
	F1 (%) ↑	98.16 ± 0.25	95.16 ± 1.84	95.65 ± 0.37	96.84 ± 0.25	96.33 ± 0.27	97.19 ± 0.44	97.38 ± 1.95	95.98 ± 1.16	98.21 ± 0.15	$\mathbf{98.87 \pm 0.07}$
	ECE (%) ↓	1.29 ± 0.17	2.86 ± 1.30	24.45 ± 0.21	2.36 ± 0.19	23.43 ± 0.28	1.98 ± 0.30	2.08 ± 1.61	3.14 ± 1.01	1.10 ± 0.17	$\mathbf{1.04 \pm 0.08}$
	NLL×100 ↓	8.01 ± 1.20	17.55 ± 7.42	36.35 ± 0.47	17.40 ± 2.62	34.96 ± 0.30	11.91 ± 1.92	17.16 ± 11.77	21.97 ± 6.99	7.15 ± 1.03	$\mathbf{4.13 \pm 0.19}$
	RC-AUC×10^3 ↓	1.76 ± 0.41	8.95 ± 5.36	4.83 ± 0.69	4.13 ± 0.49	14.53 ± 8.06	5.30 ± 1.72	5.26 ± 5.49	4.21 ± 1.64	3.75 ± 1.95	$\mathbf{0.37 \pm 0.07}$



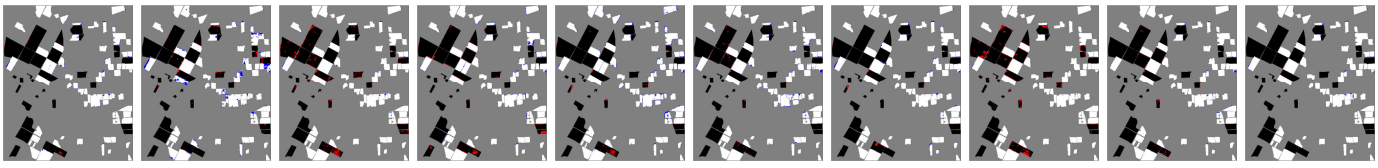
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 10. CD maps obtained by different methods on the River dataset. (a) MFCEN. (b) FDIEG. (c) DIEFEN. (d) GLAFormer. (e) AIWSEN. (f) BTSCD. (g) WCEDNet. (h) SFIEET. (i) CGMNet. (j) CCEM-Net. The FP and FN pixels are denoted in red and blue, respectively.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 11. CD maps obtained by different methods on the Farmland dataset. (a) MFCEN. (b) FDIEG. (c) DIEFEN. (d) GLAFormer. (e) AIWSEN. (f) BTSCD. (g) WCEDNet. (h) SFIEET. (i) CGMNet. (j) CCEM-Net. The FP and FN pixels are denoted in red and blue, respectively.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 12. CD maps obtained by different methods on the Bay Area dataset. (a) MFCEN. (b) FDIEG. (c) DIEFEN. (d) GLAFormer. (e) AIWSEN. (f) BTSCD. (g) WCEDNet. (h) SFIEET. (i) CGMNet. (j) CCEM-Net. The FP and FN pixels are denoted in red and blue, respectively.

Overall, CCEM-Net consistently yields cleaner and more structured change maps across all scenarios. These results align with the quantitative analysis, demonstrating improved detection accuracy and more reliable predictions.

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TABLE IV
ABLATION STUDY OF THE PROPOSED FRAMEWORK ON THREE DATASETS

No.	Ablation Components					Dataset	Classification Performance			Reliability Assessment		
	CAI	Spe.	Spa.	Yager	CCEM		OA (%) ↑	$\kappa \times 100$ ↑	F1 (%) ↑	ECE (%) ↓	NLL×100 ↓	RC-AUC×10^3 ↓
1						River	95.49 ± 0.22	68.21 ± 1.19	84.08 ± 0.60	4.79 ± 0.74	19.29 ± 1.18	24.00 ± 6.20
						Farm.	96.56 ± 0.23	91.74 ± 0.52	95.87 ± 0.26	5.07 ± 0.66	15.92 ± 1.09	13.77 ± 2.47
						BayA.	79.11 ± 2.77	57.89 ± 5.38	78.91 ± 2.63	5.85 ± 1.53	40.37 ± 2.86	51.81 ± 12.06
2	✓					River	96.54 ± 0.38	76.66 ± 3.56	88.32 ± 1.79	8.29 ± 0.38	18.17 ± 1.16	2.55 ± 0.58
						Farm.	97.62 ± 0.18	94.22 ± 0.45	97.11 ± 0.22	7.90 ± 0.33	15.35 ± 0.40	4.10 ± 1.19
						BayA.	97.72 ± 0.10	95.41 ± 0.20	97.71 ± 0.10	9.32 ± 0.46	16.17 ± 0.44	1.55 ± 0.06
3	✓	✓				River	96.90 ± 0.19	79.46 ± 1.84	89.72 ± 0.92	10.63 ± 0.52	19.87 ± 0.86	1.99 ± 0.18
						Farm.	97.64 ± 0.23	94.28 ± 0.57	97.14 ± 0.29	8.53 ± 0.33	15.61 ± 0.30	2.97 ± 1.42
						BayA.	98.58 ± 0.07	97.16 ± 0.15	98.58 ± 0.07	10.05 ± 0.29	14.78 ± 0.45	0.70 ± 0.07
4	✓	✓	✓			River	97.32 ± 0.06	82.38 ± 0.53	91.19 ± 0.27	13.34 ± 0.50	20.86 ± 0.66	1.78 ± 0.21
						Farm.	98.07 ± 0.15	95.32 ± 0.37	97.66 ± 0.18	10.07 ± 0.35	15.48 ± 0.27	$\mathbf{1.10 \pm 0.23}$
						BayA.	98.85 ± 0.06	97.69 ± 0.11	98.85 ± 0.06	10.77 ± 0.24	14.35 ± 0.25	0.42 ± 0.08
5	✓	✓	✓	✓		River	97.23 ± 0.22	81.38 ± 2.31	90.68 ± 1.17	4.91 ± 0.92	12.38 ± 1.49	1.90 ± 0.34
						Farm.	98.08 ± 0.11	95.33 ± 0.29	97.67 ± 0.15	2.78 ± 0.03	8.05 ± 0.21	1.31 ± 0.27
						BayA.	98.87 ± 0.07	97.72 ± 0.14	98.86 ± 0.07	3.15 ± 0.14	6.20 ± 0.18	0.39 ± 0.08
6	✓	✓	✓	✓	✓	River	$\mathbf{97.36 \pm 0.05}$	$\mathbf{82.71 \pm 0.40}$	$\mathbf{91.35 \pm 0.20}$	$\mathbf{1.64 \pm 0.25}$	$\mathbf{8.70 \pm 0.41}$	$\mathbf{1.78 \pm 0.12}$
						Farm.	$\mathbf{98.10 \pm 0.08}$	$\mathbf{95.40 \pm 0.19}$	$\mathbf{97.70 \pm 0.10}$	$\mathbf{0.76 \pm 0.09}$	$\mathbf{6.27 \pm 0.27}$	1.27 ± 0.17
						BayA.	$\mathbf{98.88 \pm 0.07}$	$\mathbf{97.75 \pm 0.15}$	$\mathbf{98.87 \pm 0.07}$	$\mathbf{1.04 \pm 0.08}$	$\mathbf{4.13 \pm 0.19}$	$\mathbf{0.37 \pm 0.07}$

E. Ablation Study

1) Overall Framework Analysis: We conduct a progressive ablation study to validate each component of CCEM-Net, as summarized in Table IV. All variants share the same evidential learning objective to ensure fair comparison.

Starting from the baseline (No. 1), introducing the CAI module (No. 2) yields substantial improvements across all datasets, especially on Bay Area (OA: 79.11% → 97.72%), demonstrating the effectiveness of discrepancy-guided interaction in suppressing pseudo-changes.

Further incorporating spectral and spatial branches (No. 3 and 4) consistently improves OA, Kappa, and F1, validating the benefit of decoupled multiscale representation for modeling heterogeneous patterns. Notably, on Farmland where $L = 1$, the spatial branch degenerates to per-channel modulation; yet, the decoupled design still yields consistent gains (No. 3 versus No. 4), confirming that channel-wise modulation diversity alone provides complementary information to the spectral branch. However, in No. 4, calibration degrades noticeably (e.g., ECE on River increases to 13.34%), indicating that naive fusion amplifies overconfidence under inconsistent branch predictions.

To address this issue, Yager's rule is introduced in No. 5 to explicitly handle conflicting evidence, significantly improving calibration (e.g., ECE: 10.07% → 2.78% on Farmland). Building upon this, the proposed CCEM (No. 6) further enhances both accuracy and reliability by quantifying interbranch inconsistency via JS divergence and adaptively modulating evidence before fusion. The full model achieves the best overall performance, reducing ECE to around 1% while maintaining superior accuracy.

To further illustrate this effect, Fig. 13 visualizes the predictions of the spectral branch, spatial branch, and CCEM-Net, along with the corresponding JS divergence. High-conflict regions align with inconsistent branch predictions, particularly along boundaries and small structures. By suppressing unreliable evidence in these regions, CCEM-Net produces more coherent results.

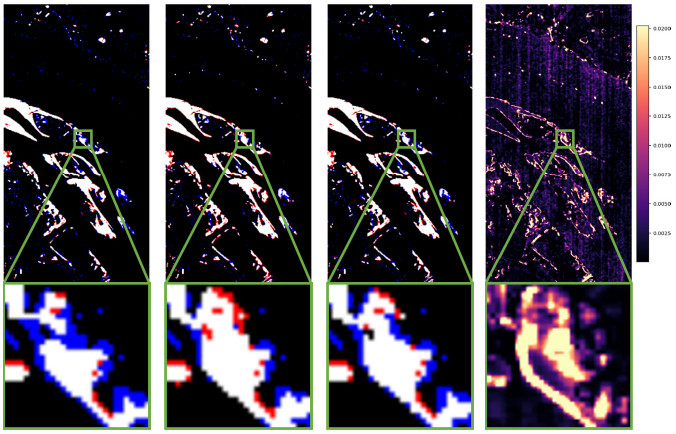


Fig. 13. Visual comparison of CD maps and the JS divergence on the River dataset. (a) Spatial branch CD map. (b) Spectral branch CD map. (c) CCEM-Net CD map. (d) JS divergence heatmap indicating interbranch conflict.

2) Analysis of the CCEM Mechanism: To further justify the design of CCEM, we conduct a fine-grained ablation by individually evaluating adaptive conflict modeling, conflict-aware discounting, determinacy weighting, the learnable temperature, and the evidential fusion rule. The results are summarized in Table V.

Replacing the proposed adaptive conflict modeling with a fixed conflict assumption ($JS = 1$) causes the largest calibration degradation on all datasets while leaving OA nearly unchanged (e.g., ECE increases from 1.68% to 4.10% on River). This indicates that estimating sample-wise interbranch conflict is crucial for reliable evidence modulation: a fixed maximum-conflict assumption causes the discount factor to indiscriminately suppress evidence even when the two branches agree, penalizing mutually consistent predictions and degrading calibration.

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TABLE V
FINE-GRAINED ABLATION STUDY OF THE PROPOSED CCEM MECHANISM

Variant	River	Farmland	Bay Area
	OA / ECE (%)	OA / ECE (%)	OA / ECE (%)
Full CCEM	**97.36 / 1.68**	**98.10 / 0.76**	**98.88 / 1.04**
Fixed conflict	97.34 / 4.10	**98.10** / 3.48	98.84 / 2.89
w/o δ_b	97.29 / 2.45	98.08 / 0.87	98.86 / 1.22
w/o d_b	97.26 / 4.04	98.08 / 2.77	98.86 / 3.01
Fixed $\tau = 1$	97.34 / 1.69	**98.10** / 0.77	**98.88** / 1.07
Dempster fusion	97.31 / 1.75	98.09 / 0.90	98.86 / 1.09

TABLE VI
COMPARISON OF DIFFERENT BRANCH FUSION STRATEGIES

Fusion Strategy	River	Farmland	Bay Area
	OA / ECE (%)	OA / ECE (%)	OA / ECE (%)
Average	97.31 / 13.31	98.05 / 9.94	98.86 / 10.64
Concat+MLP	97.17 / 10.49	**98.12** / 6.53	98.92 / 5.87
Gated Fusion	97.33 / 11.62	98.11 / 8.03	**98.96** / 10.25
Cross Attention	97.35 / 8.01	97.96 / 8.36	98.92 / 8.35
CCEM	**97.36 / 1.68**	98.10 / **0.76**	98.88 / **1.04**

Removing either the conflict-aware discount factor δ_b or the determinacy weighting d_b also degrades calibration, demonstrating that branch reliability should jointly consider interbranch disagreement and predictive uncertainty. In contrast, fixing the learnable temperature ($\tau = 1$) results in only marginal performance changes, suggesting that the proposed weighting strategy is inherently robust. Replacing Yager's rule with Dempster's rule consistently produces slightly higher ECE, confirming that explicitly transferring conflicting evidence into uncertainty yields more reliable evidential fusion.

Overall, these results verify that adaptive conflict estimation, reliability-aware evidence modulation, and Yager-rule-based evidential fusion jointly contribute to the superior calibration of CCEM while preserving high detection accuracy.

3) Comparison With Different Fusion Strategies: To further validate the effectiveness of CCEM, we replace the proposed conflict-aware evidential fusion with several representative fusion strategies while keeping the same backbone and evidential prediction heads. Specifically, average fusion, Concat+MLP, gated fusion, and cross attention are adopted as alternative branch fusion strategies. The comparison results are summarized in Table VI.

As shown in Table VI, all competing methods achieve comparable OA across the three datasets, indicating that the proposed spectral–spatial backbone already provides sufficiently discriminative representations. However, significant differences are observed in calibration performance. Average fusion consistently produces the highest ECE because it assigns equal importance to both branches regardless of their reliability. Concat+MLP and Gated Fusion introduce learnable fusion weights and achieve improved calibration; yet, they still fail to explicitly model predictive uncertainty or interbranch conflict. Cross Attention further enhances feature interaction and obtains lower ECE on the River dataset, but its calibration performance remains inconsistent across datasets.

In contrast, CCEM consistently achieves the lowest ECE while maintaining competitive OA. This demonstrates that the performance gain does not simply originate from employing a more sophisticated fusion architecture. Instead, explicitly modeling branch evidence, estimating interbranch conflict, and performing reliability-aware evidential fusion are critical for obtaining reliable predictions.

TABLE VII
COMPARISON OF STANDARD EDL AND RE-EDL LOSS FUNCTIONS

Loss Function	River	Farmland	Bay Area
	OA / ECE (%)	OA / ECE (%)	OA / ECE (%)
Re-EDL	97.32 / 2.16	98.04 / 1.15	98.87 / 1.65
Standard EDL	**97.36 / 1.68**	**98.10 / 0.76**	**98.88 / 1.04**

4) Analysis of the Evidential Loss Function: The training objective in (24)–(28) follows the standard EDL formulation [30], while Re-EDL [44] removes the variance term and KL regularization to alleviate potential over- and under-confidence. To investigate whether the Yager-rule-based evidential fusion changes the behavior of these two terms, we replace only the evidential loss with the Re-EDL objective while keeping all other components unchanged. The results are summarized in Table VII.

As shown in Table VII, the standard EDL objective consistently achieves higher OA and lower ECE than Re-EDL across all three datasets, indicating that the original variance and KL terms remain beneficial under the proposed framework.

We attribute this improvement to the interaction between the standard EDL objective and the proposed conflict-aware evidential fusion. Since unreliable evidence is suppressed before fusion and conflicting mass is explicitly transferred into uncertainty by Yager's rule, the variance term mainly regularizes the remaining overconfident predictions rather than indiscriminately penalizing high-evidence samples. Meanwhile, the annealed KL regularization helps to maintain a balanced evidence distribution during training. Consequently, the adverse effects reported by Re-EDL for conventional single-branch EDL are largely alleviated, making the standard EDL objective more suitable for the proposed CCEM framework.

F. Generalization Evaluation Under Different Training Protocols

To further evaluate the generalization ability of CCEM-Net beyond the standard 1%/99% random split protocol, we conduct two complementary experiments. First, we evaluate performance under different training ratios (0.5%, 1%, 2%, 5%, and 10%) to assess data efficiency. Second, we adopt a spatial split protocol to eliminate the neighborhood overlap introduced by random patch-based sampling, thereby providing a more rigorous evaluation of spatial generalization.

As shown in Fig. 14, CCEM-Net consistently achieves the highest OA and the lowest ECE across all training ratios on the three datasets. The advantage is particularly evident under

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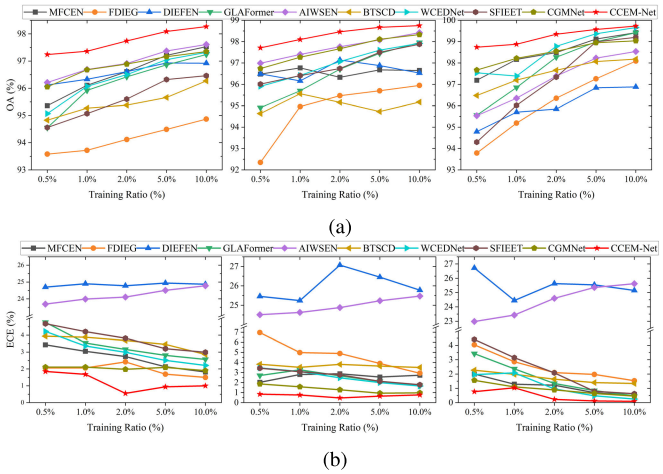


Fig. 14. Performance trends of all methods under varying training ratios (0.5%, 1%, 2%, 5%, and 10%) across three datasets. (a) OA versus training ratio. (b) ECE versus training ratio.

TABLE VIII

COMPARISON OF SPATIAL GENERALIZATION UNDER TOP–BOTTOM SPLIT PROTOCOL (OA% ↑ / ECE% ↓)

Method	River		Farmland		Bay Area	
	T-B	B-T	T-B	B-T	T-B	B-T
MFCEN	93.29 / 5.08	97.51 / 1.83	94.22 / 5.26	98.50 / 1.31	90.01 / 6.81	86.05 / 11.90
FDIEG	88.51 / 8.14	92.91 / 4.25	92.21 / 2.88	95.60 / 2.79	86.96 / 8.33	78.17 / 16.08
DIEFEN	94.22 / 22.94	97.39 / 25.61	94.54 / 23.24	98.63 / 26.84	88.75 / 19.13	89.44 / 19.90
GLAFormer	91.41 / 7.87	96.81 / 2.84	94.00 / 5.18	98.23 / 1.52	94.15 / 5.46	93.08 / 5.95
AIWSEN	92.39 / 21.55	97.84 / 25.99	94.25 / 23.46	98.85 / 26.78	92.91 / 21.80	93.11 / 22.15
BTSCD	92.54 / 5.51	96.00 / 2.75	93.96 / 3.85	95.95 / 2.94	89.95 / 6.04	86.92 / 8.44
WCEDNet	92.85 / 4.74	97.38 / 1.90	92.14 / 5.86	98.41 / 1.08	92.82 / 3.70	93.35 / 5.34
SFIEET	92.05 / 7.28	96.13 / 3.38	92.53 / 6.60	98.38 / 1.41	92.39 / 6.48	90.52 / 8.05
CGMNet	92.74 / 5.80	97.31 / 2.13	93.67 / 5.15	98.54 / 1.06	94.89 / 4.17	95.86 / 3.19
CCEM-Net	**95.11 / 3.18**	**98.42 / 1.11**	**96.75 / 2.24**	**99.00 / 0.65**	**97.94 / 1.07**	**97.48 / 0.56**

extremely limited supervision (0.5% and 1%), demonstrating the superior data efficiency of the proposed evidential conflict modeling. Although the accuracy of all methods improves with increasing training data, several softmax-based approaches (e.g., DIEFEN and AIWSEN) remain severely overconfident, with ECE consistently exceeding 20%. In contrast, CCEM-Net maintains ECE below approximately 2% across all datasets, indicating that its reliability stems from uncertainty-aware evidential fusion rather than simply benefiting from more labeled samples.

To further address the potential spatial overlap caused by random patch-based sampling, we additionally adopt a top–bottom spatial split protocol. Specifically, each dataset is divided into two spatially disjoint halves. Training is performed on one half (2% random sampling within that half, approximately equivalent to 1% of the entire scene), while testing is conducted on the other half. The two directions are denoted by T-B and B-T.

Table VIII shows that all methods experience performance degradation under the spatial split protocol, confirming that random patch-based sampling indeed overestimates performance to some extent. Nevertheless, CCEM-Net consistently achieves the best OA and the lowest ECE across all datasets and both split directions. For example, on the River dataset, the OA decreases from 97.36% under random sampling to 95.11% under the more challenging T-B split, whereas several competing methods exhibit substantially larger drops. Moreover, methods trained with conventional softmax loss (e.g., DIEFEN and AIWSEN) remain highly overconfident, with ECE exceeding 20% regardless of the split direction. These results demonstrate that the proposed conflict-aware evidential fusion generalizes well to spatially disjoint regions and is not merely benefiting from neighborhood overlap in the standard random split protocol.

G. Uncertainty and Reliability Analysis

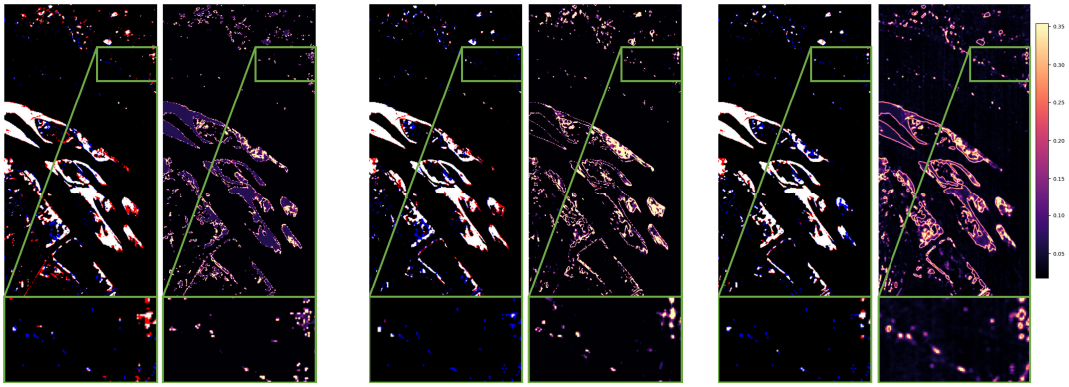
To comprehensively evaluate the reliability of CCEM-Net, we analyze its uncertainty estimation from four perspectives: visual inspection, classwise calibration, reliability diagram analysis, and decision-level risk control.

First, we examine failure modes of uncertainty estimation through visual inspection of error maps and uncertainty heatmaps. Figs. 15–17 present these comparisons for CCEM-Net alongside MFCEN and CGMNet—the two baselines that achieve the strongest overall calibration. Two patterns of mismatch between uncertainty and prediction correctness may be identified from these visualizations.

The first pattern—overconfident errors where misclassified pixels coincide with low uncertainty—can be observed in the MFCEN and CGMNet panels across all three datasets. Most of these errors occur around object boundaries and transition regions, where spectrally ambiguous pixels are assigned high confidence despite being incorrectly classified. In contrast, CCEM-Net assigns elevated uncertainty to many of these regions, and the majority of its residual errors are located within or near high-uncertainty areas. This behavior is consistent with the proposed conflict-aware fusion mechanism: when spectral and spatial evidence disagree, the elevated interbranch conflict suppresses overconfident evidence and transfers the conflicting mass into predictive uncertainty through Yager-rule fusion.

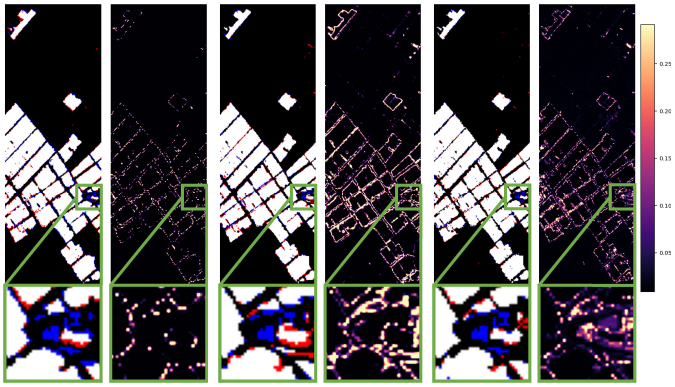
The second pattern—conservatively elevated uncertainty on correctly classified pixels—may be observed in the CCEM-Net panels. On the River dataset, certain unchanged pixels adjacent to the watercourse that are correctly classified nevertheless exhibit moderate uncertainty in the heatmap. A similar behavior can be seen in Farmland, where pixels near field boundaries, as well as in Bay Area, where pixels along urban edges, retain nonnegligible uncertainty despite being correctly classified. This behavior is likely related to the evidential representation adopted by CCEM-Net: when the spectral and spatial branches provide only moderately consistent evidence, the fused Dirichlet distribution may retain a nonnegligible degree of uncertainty rather than collapsing to highly confident predictions. While this may slightly increase uncertainty on some correctly classified samples, the asymmetry in failure cost—a conservative prediction on an unchanged pixel is typically less consequential than an overconfident error on a changed pixel—suggests that this tradeoff may be acceptable for HSCD. The classwise calibration analysis presented next provides quantitative evidence that this conservative behavior does not substantially compromise Unchanged-class calibration, while the changed class—the

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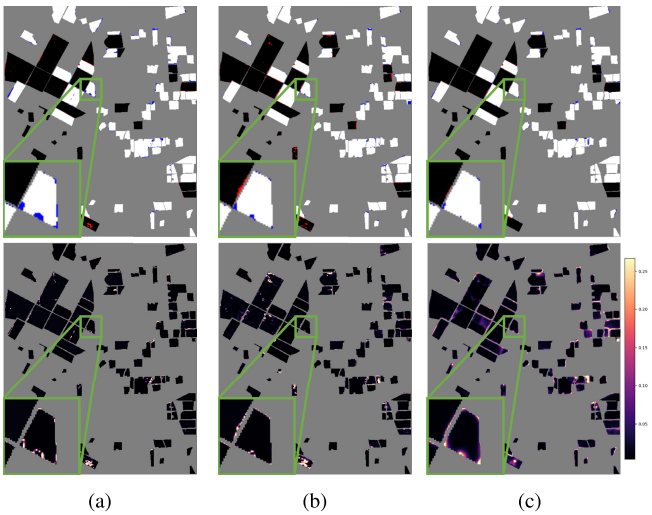
(a) (b) (c)

Fig. 15. Visual comparison of CD maps and uncertainty heatmaps on the River dataset. (a) MFCEN. (b) CGMNet. (c) CCEM-Net.



(a) (b) (c)

Fig. 16. Visual comparison of CD maps and uncertainty heatmaps on the Farmland dataset. (a) MFCEN. (b) CGMNet. (c) CCEM-Net.



(a) (b) (c)

Fig. 17. Visual comparison of CD maps and uncertainty heatmaps on the Bay Area dataset. (a) MFCEN. (b) CGMNet. (c) CCEM-Net.

primary target in HSCD—benefits from markedly improved reliability.

Second, to examine whether the calibration improvement is consistent across changed and unchanged classes, we report classwise ECE for CCEM-Net and three representative softmax-based baselines: AIWSEN (poorly calibrated), together with MFCEN and CGMNet (stronger calibration baselines). The results are summarized in Table IX, where the class proportions are also reported.

TABLE IX
CLASSWISE CALIBRATION COMPARISON (ECE%, CHANGED/UNCHANGED)

Method	River (8.69% / 91.31%)	Farmland (29.01% / 70.99%)	Bay Area (53.44% / 46.56%)
MFCEN	15.87 / 1.84	4.37 / 2.10	1.76 / **0.75**
AIWSEN	8.03 / 25.43	24.02 / 24.90	23.12 / 24.08
CGMNet	15.86 / **0.81**	1.87 / 1.46	1.41 / **0.75**
CCEM-Net	**5.22** / 1.96	**0.94** / **0.78**	**0.78** / 1.39

As shown in Table IX, the representative baselines exhibit two typical calibration failure modes. AIWSEN exhibits high ECE on both classes, with particularly severe miscalibration on the unchanged class, suggesting that its predicted confidence is poorly aligned with prediction correctness. MFCEN and CGMNet achieve low unchanged ECE but substantially higher changed ECE, indicating that softmax-based confidence may be reasonably calibrated on homogeneous unchanged regions yet poorly aligned with correctness on changed pixels, where spectral ambiguity and boundary mixing are more prevalent. In contrast, CCEM-Net achieves consistently low changed ECE across all three datasets. The slightly higher unchanged ECE on River and Bay Area likely reflects the conservative nature of evidential modeling: by representing predictions as Dirichlet distributions, the model retains a modest degree of uncertainty even on easy-to-classify pixels. Overall, the classwise analysis confirms that CCEM-Net provides well-calibrated predictions on the changed class—the primary target in HSCD—with no substantial degradation on the unchanged class.

Third, we evaluate calibration quality using Reliability Diagrams on the River dataset (see Fig. 18), which is more challenging due to complex boundary structures and spectral variability. The red shaded regions indicate the gap between predicted confidence and empirical accuracy. Baseline methods, particularly AIWSEN (ECE = 23.91%), exhibit severe overconfidence, while MFCEN and CGMNet still show

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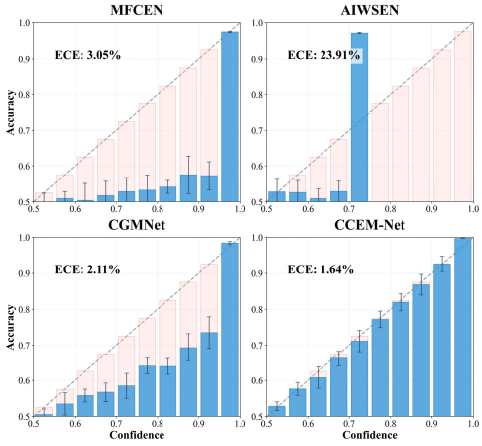


Fig. 18. Reliability diagrams of four different methods on the River dataset.

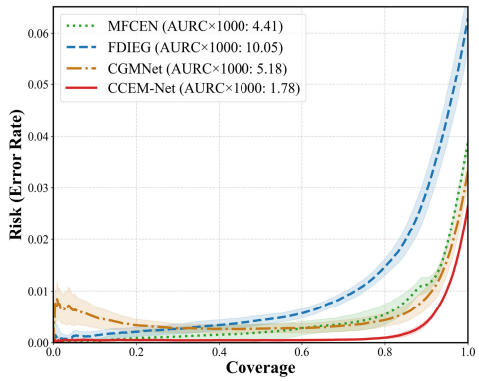


Fig. 19. RC curves of four methods on the River dataset.

noticeable calibration gaps. In comparison, CCEM-Net closely follows the ideal diagonal line and achieves the lowest ECE (1.64%), demonstrating well-calibrated confidence estimates.

Finally, we assess the practical utility of uncertainty through RC curves (see Fig. 19), also on the River dataset. A lower curve indicates better selective prediction performance. CCEM-Net consistently maintains lower risk across all coverage levels and achieves the smallest RC-AUC (e.g., 1.78×10^{-3}). As coverage decreases, its risk drops rapidly, indicating that uncertain samples can be effectively identified and filtered, which is particularly valuable in risk-sensitive scenarios.

Overall, these results demonstrate that CCEM-Net achieves reliable uncertainty estimation through explicit evidential modeling and conflict-aware modulation. By jointly modeling predictive uncertainty and interbranch inconsistency, the proposed framework produces well-calibrated and trustworthy predictions.

H. Complexity and Efficiency Analysis

Table X presents a quantitative comparison of the proposed CCEM-Net against several SOTA methods on the River dataset in terms of model complexity and execution efficiency. It can be observed that CCEM-Net achieves superior performance across all evaluation metrics.

Regarding model parameters and computational cost, CCEM-Net demonstrates a significant reduction in overhead. Specifically, the parameters of CCEM-Net are only 0.22 M, which is approximately 15% of the second-best method (WCEDNet, 1.42M) and less than 0.15% of the traditional MFCEN (153.91M). Similarly, its FLOPs are a mere 0.13 G, which is over two orders of magnitude lower than Transformer-based approaches like GLAFormer (62.51G). This exceptional efficiency is primarily attributed to our decoupled AMSS module, which utilizes depthwise separable multiscale convolutions to eliminate redundant computations while preserving robust feature extraction capabilities.

TABLE X
COMPARISON OF COMPUTATIONAL COMPLEXITY AND EXECUTION EFFICIENCY ON THE RIVER DATASET

Methods	Par. (M) ↓	FLOPs (G) ↓	Tr. (s) ↓	Te. (s) ↓
MFCEN	153.91	16.26	296.77	45.56
FDIEG	2.18	8.78	160.47	31.69
DIEFEN	9.61	1.43	241.59	38.82
GLAFormer	12.67	62.51	284.22	43.25
AIWSEN	12.51	5.43	284.22	43.25
BTSCD	25.50	3.82	211.39	35.25
WCEDNet	1.42	19.16	211.24	46.89
SFIEET	1.89	10.50	161.53	31.13
CGMNet	15.33	11.33	330.75	56.97
CCEM-Net	**0.22**	**0.13**	**157.12**	**17.22**

Note: Par. denotes trainable parameters; FLOPs are computed per forward pass; Tr. and Te. refer to the total training and testing time, respectively.

In terms of execution time, CCEM-Net maintains the highest inference speed at 17.22 s. Although the CCEM mechanism is introduced for reliability, it involves primarily closed-form operations, thereby avoiding any substantial computational bottleneck or iterative optimization during inference.

In summary, these results demonstrate that CCEM-Net achieves an exemplary balance among detection accuracy, predictive reliability, and computational efficiency, making it highly suitable for practical HSCD in resource-constrained scenarios.

V. CONCLUSION

In this article, we propose CCEM-Net, a conflict-aware EDL framework for HSCD. By jointly modeling cross-temporal dependencies, multiscale spectral–spatial representations, and conflict-aware decision fusion within a unified architecture, the proposed method effectively addresses key challenges in HSCD, including spectral inconsistency, heterogeneous spatial patterns, and ambiguous decision boundaries.

In particular, the proposed CCEM mechanism introduces evidential modeling to explicitly quantify predictive uncertainty and resolve interbranch conflicts, providing a principled approach for reliable decision-level fusion under ambiguous conditions. Experimental results on multiple benchmark datasets demonstrate that CCEM-Net achieves competitive performance in terms of both accuracy and reliability. Furthermore, uncertainty analysis confirms its ability to capture predictive ambiguity and mitigate the influence of conflicting evidence.

Despite the promising results, several limitations of the proposed framework should be noted. First, CCEM-Net relies

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on pixel-level annotations for supervised training. Although the training ratio analysis in Section IV-F demonstrates robust performance down to 0.5% labeled samples, its accuracy under extremely scarce annotations may still be insufficient for applications where labeled data is severely limited, and the framework does not yet support fully unsupervised deployment. Second, the patch size is currently selected on a per-dataset basis; its optimal value may vary with the spatial resolution and land-cover characteristics of different hyperspectral sensors, requiring empirical tuning when applied to new scenes. Third, the framework assumes that bitemporal HSIs are accurately co-registered. In the presence of significant misregistration, the pixelwise comparison between spectral and spatial branches may degrade, and the interbranch conflict estimated by JS divergence may conflate genuine surface changes with geometric misalignment artifacts. These limitations point to important directions for future investigation.

Future work will explore extending the proposed framework to more challenging scenarios, including large-scale cross-scene change detection and weakly supervised settings, as well as developing more efficient evidential modeling strategies for practical applications.

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